

Question Number	Answer	Additional Guidance	Mark
4(a)	An answer that includes three of the following: • find a suitable {concentration / volume} of pineapple juice (1) • find a suitable timescale for growth of bacteria (1) • find a suitable {temperature / pH} for {growth of bacteria / incubation / activity of enzyme} (1) • find a suitable way to count bacteria (1)	Accept find a suitable range of concentrations Accept find a suitable method of making the pineapple juice Accept find storage time (of pineapple juice) which has an effect Accept identify suitable growth medium Accept find a suitable method for comparing growth (rate)	(3)

Question Number	Answer	Additional Guidance	Mark
4(b)	An answer that includes eight of the following: • clear statement of the dependent variable (1) • method of producing pineapple juice (1) • (use of nutrient broth) with and without pineapple juice (1) • use of aseptic technique (1) • samples taken at suitable intervals (1) • suitable method for counting bacteria (1) • method of calculation of growth rate (1) • identify one variable to be controlled and description of how it is controlled (1) • identify second variable to be controlled and description of how it is controlled (1) • repeats for each set up or repeat the whole investigation (1)	Accept {number of bacteria / turbidity / absorbance} with ref to time Accept heated juice / water / sugar solution as a control Accept several concentrations of juice as long one is zero. Accept comparison of fresh and stored eg flaming neck of bottles, work near Bunsen burner etc Accept bacteria counted at start and after suitable time Accept {absorbance / transmission} with {colorimeter / datalogger} Accept serial dilution and plating Accept: temperature, pH, {age / variety / part / tissue} of pineapple, {concentration / volume} of juice, {age / volume / concentration} of bacterial culture {composition / concentration / volume} of growth medium	(8)

Question Number	Answer	Additional Guidance	Mark
4(c)	An answer that includes the following: • raw data table with headings and appropriate units, with means calculated from repeats (1) • graph with both labelled axes (1) • use of an appropriate statistical test (1)	Accept description of calculating mean	(3)

Question Number	Answer	Additional Guidance	Mark
4(d)	An answer that includes three of the following: • {clumping / uneven distribution} of bacteria makes it difficult to count (1) • factors related to colorimeter use (1) • relevant named factor may affect growth (rate) of bacteria (1) • second relevant named factor may affect growth (rate) of bacteria (1)	Accept hard to add the same number of bacteria to each culture at the start Accept factors related to serial dilution and plating (1) Ignore pH and temperature	(3)

(Total for question 4 = 17 marks)

